

vidual source is pre-rated according to the limit applicable to its frequency.

For simultaneous exposure to fields at different frequencies of interest (GSM, CDMA, HSDPA, WiMAX), compliance with the exposure limits is evaluated by considering field strengths due to all the frequencies of interest.

Spatial averaging. Strength of EM fields of an antenna is highly dependent on the distance/position from the antenna. Near close vicinity of the antenna, these fields are appreciably high. So, spatial averaging is required to get an exact estimation of field strength. Moreover, field strength may also vary with the spatial position due to the effect of reflection, shadowing and scattering about adjacent conducting surfaces. Fig. 3 clarifies such a situation where reflected fields add more values to the fields that may lead to over-estimation of the field during measurements.

Multi-path reflections can create non-uniform field distribution. Therefore to assess the correct exposure levels, an averaging process is required. Field values should be determined at several points (p) as shown in Fig. 4. Basically, three points are recommended (Fig. 4a), but if higher accuracy is required, the number can be increased to six (Fig. 4b), nine (Fig. 4c), 20 (Fig. 4d) and so on. If there are multiple sources, measurement is performed by dividing the entire area into small

grids of about one-square-metre each and taking measurements at individual grid points.

Source power and radiation pattern variability. In telecommunication systems, transmitted power may vary in accordance with the usage of channel (channel allocation) and power requirements in certain atmospheric conditions. Some channels are turned on or off as per need basis in dynamic channel allocation. Variation in channel occupancy (where amount of data transmitted over a channel varies; even if no data is transmitted, channel carrier remains) also alters the amount of transmitted power.

To compensate for adverse propagation conditions, the base station uses automatic power control that results in variation of transmitted power. Sometimes, advanced telecom systems use active antennae that can dynamically change their radiation patterns. These variabilities impose a greater challenge for measurements, since the exact state of the transmitter at the time of measurement may not be known.

Assessment of radiation levels

Assessment of exposure levels can be done either by measurement, by numerical calculations or by EM software simulations. All these methods have almost similar levels of uncertainty and accuracy depending on the method, equipment or software used.

Assessment by measurement.

Broadband measurement. An isotropic field probe may be used for broadband measurement of electric field intensity (V/m) or power density (watt/sqm). For a normally compliant radio-communication site, overall measured value of E field or power density with broadband measurement

should be within 10 per cent of the reference levels prescribed by ICNIRP for occupational/general public. With broadband measurement, the overall picture of radiation is obtained, but it is hard to get information about contributions to overall radiation level made by individual sources such as GSM, UMTS, CDMA, WiMAX or any other radio-communication system (AM/FM/TV).

A typical test setup for E field measurement is shown in Fig. 5. Some broadband antennae like bi-conical, log-periodic or ridge-guide may be used for measurement in the desired radio-communication frequency band (GSM 900/1800, HSDPA, WiMAX, etc). In such type of antennae, it is required to align the antenna in the plane of incident E field and, thus, the knowledge of polarisation of the radiation source is required.

Another type of antenna/probe that is increasingly being favoured for use with the spectrum analyser is the tri-axial isotropic antenna/probe. This is made by three independent broadband sensing elements placed orthogonally to each other. It simplifies the measurement procedure that ideally determines RMS field strength independent of direction and polarisation of the radiation source, since it affords measurement of E field components along all three axes (X, Y and Z).

Measured value of E_x , E_y and E_z , the electrical field strengths corresponding to each axis of the tri-axial isotropic probe, are first corrected for their antennae factors (for each axis) as:

$$E = AF \times V$$

where E is the electric field strength in volts/metre, V is voltage received by the spectrum analyser and AF is the antenna factor.

Unit of AF is 1/metre. Antenna factor is antenna specific and normally supplied by the manufacturer. Now, the resultant E field strength is added geometrically (root sum of squares).

Frequency selective measurement. As described above, broadband measurement facilitates measurement of the combined effect of all radiating sources. To measure individual response of each and every radiating source, very high sensitive frequency

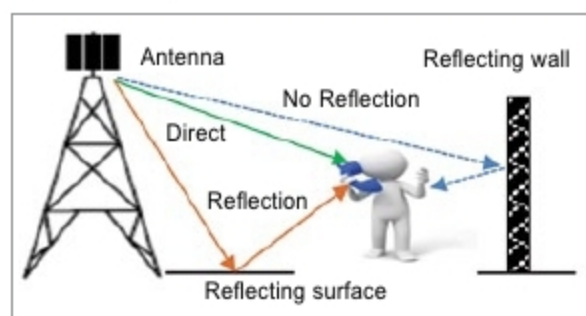


Fig. 3: Multi-path reflection of waves

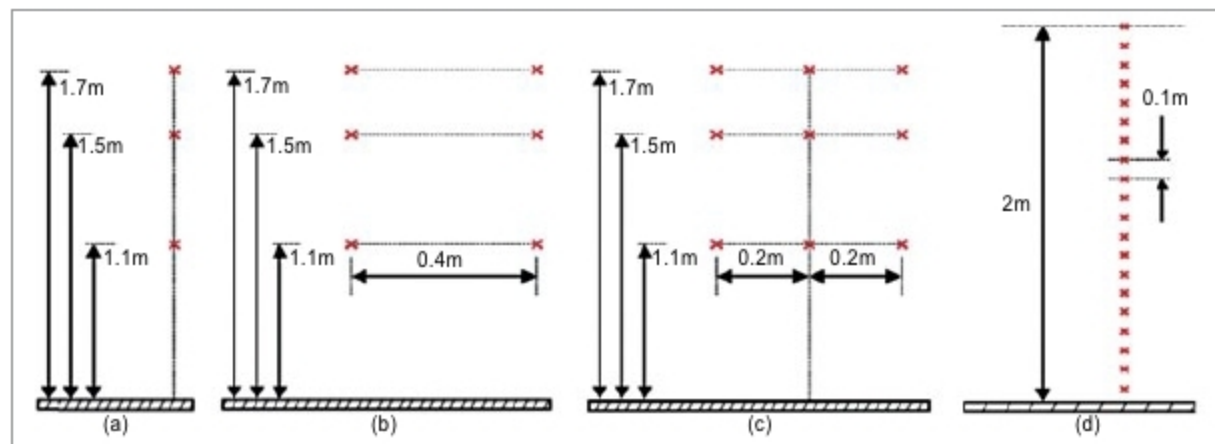


Fig. 4: Measurement points for spatial averaging